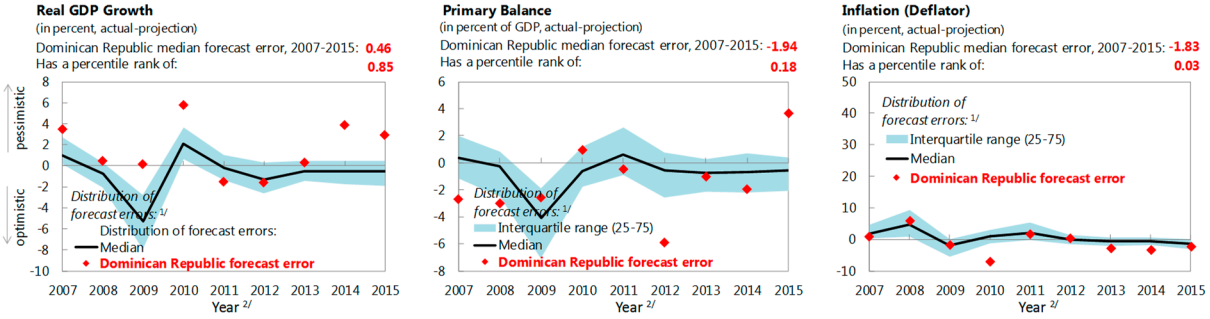
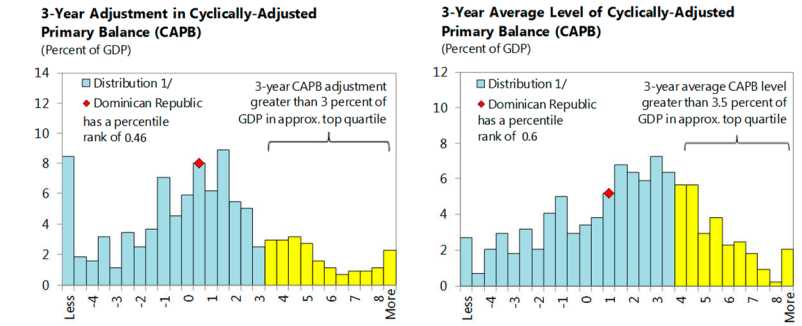
Annex III. Figure 2. Dominican Republic DSA – Realism of Baseline Assumptions

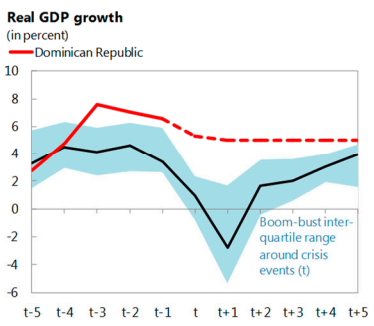
Forecast Track Record, versus all countries



Assessing the Realism of Projected Fiscal Adjustment



Boom-Bust Analysis 3/



Source: Fund staff calculations and estimates.
1/ Plotted distribution includes all countries, percentile rank refers to all countries.
2/ Projections made in the spring WEO vintage of the preceding year.
3/ For Dominican Republic, t corresponds to 2017; for the distribution, t corresponds to the first year of the crisis.
4/ Data cover annual obervations from 1990 to 2011 for advanced and emerging economies with debt greater than 60 percent of GDP. Percent of sample on vertical axis.

Annex III. Figure 3. Dominican Republic Public Sector Debt Sustainability Analysis (DSA) – Baseline Scenario

(In percent of GDP, unless otherwise indicated)

Debt, Economic and Market Indicators [1/]

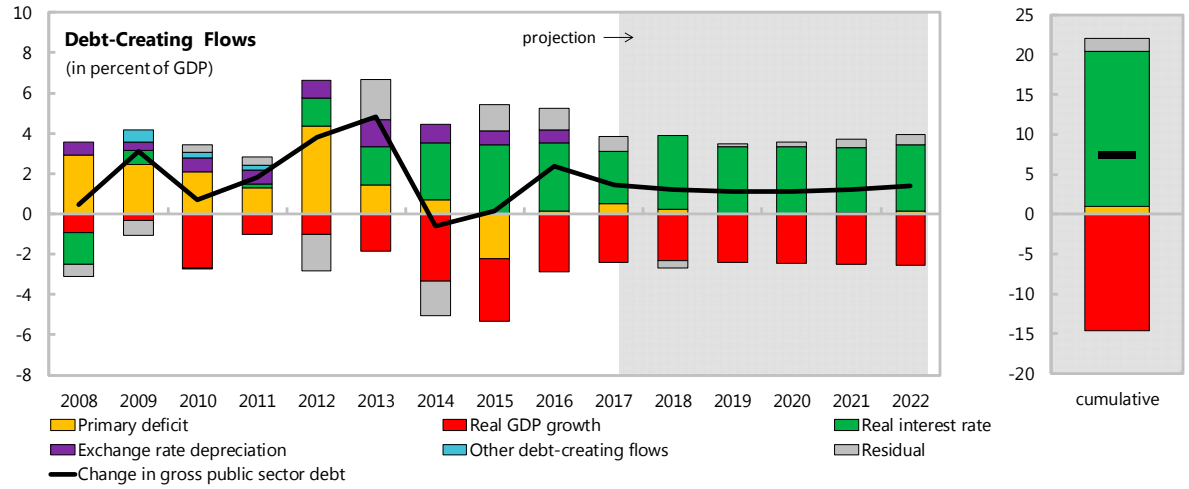
	Actual			Projections					
	2006-2014	2015	2016	2017	2018	2019	2020	2021	2022
Nominal gross public debt	39.3	47.3	49.7	51.1	52.3	53.4	54.5	55.7	57.1
Public gross financing needs	8.9	5.3	6.3	8.0	8.7	9.5	9.5	9.5	11.5
Real GDP growth (in percent)	5.5	7.0	6.6	5.3	5.0	5.0	5.0	5.0	5.0
Inflation (GDP deflator, in percent)	5.5	0.9	1.6	3.9	4.2	4.0	4.0	4.0	4.0
Nominal GDP growth (in percent)	11.3	8.0	8.3	9.4	9.4	9.2	9.2	9.2	9.2
Effective interest rate (in percent) [2/]	6.8	8.8	9.5	9.7	10.1	10.8	10.8	10.1	9.8

As of February 07, 2017

Sovereign Spreads		
EMBI (bp)		360
Ratings	Foreign	Local
Moody's	B1	B1
S&Ps	BB-	BB-
Fitch	BB-	BB-

Contribution to Changes in Public Debt

	Actual			Projections							
	2006-2014	2015	2016	2017	2018	2019	2020	2021	2022	Cumulative	Debt-stabilizing primary balance [7/]
Change in gross public sector debt	1.0	0.1	2.4	1.4	1.2	1.1	1.1	1.2	1.4	7.4	
Identified debt-creating flows	1.2	-1.2	1.3	0.7	1.6	1.0	0.9	0.8	0.9	5.8	1.6
Primary deficit	1.9	-2.2	0.2	0.5	0.2	0.0	0.0	0.1	0.2	1.0	
Primary (noninterest) revenue and grants	14.3	17.5	14.6	14.8	14.7	14.7	14.6	14.6	14.5	87.9	
Primary (noninterest) expenditure	16.2	15.3	14.8	15.3	15.0	14.7	14.6	14.6	14.7	88.9	
Automatic debt dynamics [3/]	-0.9	1.1	1.1	0.2	1.3	1.0	0.9	0.7	0.7	4.8	
Interest rate/growth differential [4/]	-1.4	0.3	0.5	0.2	1.3	1.0	0.9	0.7	0.7	4.8	
Of which: real interest rate	0.5	3.4	3.4	2.6	3.7	3.4	3.3	3.2	3.3	19.4	
Of which: real GDP growth	-1.9	-3.1	-2.9	-2.4	-2.3	-2.4	-2.4	-2.5	-2.6	-14.6	
Exchange rate depreciation [5/]	0.6	0.7	0.6	...	...	...	...	...	...	...	
Other identified debt-creating flows	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Contingent liabilities	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Floating debt	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Residual, including asset changes [6/]	-0.2	1.3	1.1	0.7	-0.4	0.1	0.2	0.4	0.5	1.6	



Source: Fund staff calculations and estimates.

1/ Public sector is defined as consolidated public sector.

2/ Defined as interest payments divided by debt stock at the end of previous year.

3/ Derived as [(r - p(1+g) - g + ae(1+r)]/(1+g+p+gp)) times previous period debt ratio, with r = interest rate; p = growth rate of GDP deflator; g = real GDP growth rate; a = share of foreign-currency denominated debt; and e = nominal exchange rate depreciation (measured by increase in local currency value of U.S. dollar).

4/ The real interest rate contribution is derived from the denominator in footnote 4 as r - π (1+g) and the real growth contribution as -g.

5/ The exchange rate contribution is derived from the numerator in footnote 2/ as ae(1+r).

6/ For projections, this line includes exchange rate changes during the projection period.

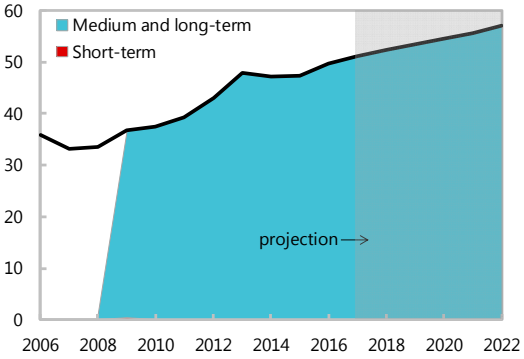
7/ Assumes that key variables (real GDP growth, real interest rate, and other identified debt-creating flows) remain at the level of the last projection year.

Annex III. Figure 4. Dominican Republic Public DSA – Composition of Public Debt and Alternative Scenarios

Composition of Public Debt

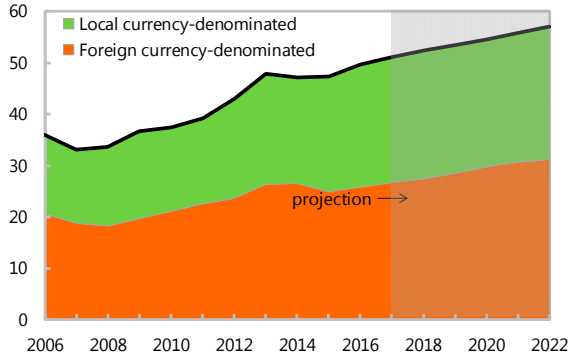
By Maturity 1/

(in percent of GDP)



By Currency

(in percent of GDP)

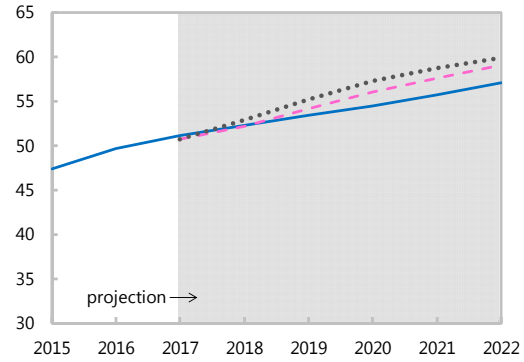


Alternative Scenarios



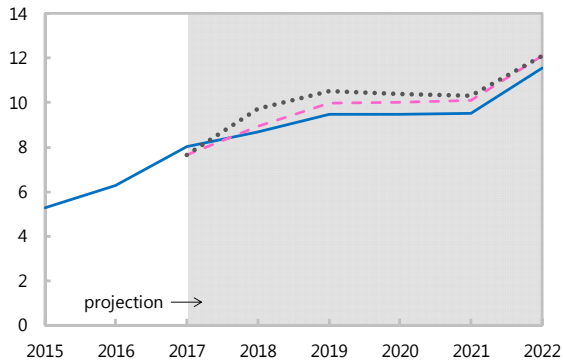
Gross Nominal Public Debt

(in percent of GDP)



Public Gross Financing Needs

(in percent of GDP)



Underlying Assumptions

(in percent)

Baseline Scenario	2017	2018	2019	2020	2021	2022
Real GDP growth	5.3	5.0	5.0	5.0	5.0	5.0
Inflation	3.9	4.2	4.0	4.0	4.0	4.0
Primary Balance	-0.5	-0.2	0.0	0.0	-0.1	-0.2
Effective interest rate	9.7	10.1	10.8	10.8	10.1	9.8
Constant Primary Balance Scenario						
Real GDP growth	5.3	5.0	5.0	5.0	5.0	5.0
Inflation	3.9	4.2	4.0	4.0	4.0	4.0
Primary Balance	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5
Effective interest rate	9.7	10.1	10.8	10.7	10.0	9.7

Historical Scenario	2017	2018	2019	2020	2021	2022
Real GDP growth	5.3	5.3	5.3	5.3	5.3	5.3
Inflation	3.9	4.2	4.0	4.0	4.0	4.0
Primary Balance	-0.5	-1.3	-1.3	-1.3	-1.3	-1.3
Effective interest rate	9.7	10.1	10.2	9.6	8.5	7.9

Source: Fund staff calculations and estimates.

1/ Short-term debt includes unidentified financing during the projection horizon.